ASSIGNMENT 2
MECHENG 4AI3: Applied Artificial Intelligence
McMaster University, Winter 2026

Instructor: Dr. Ryan Ahmed
Assignment Due Date: March 10th, 2026, at 12:00 Midnight

Problem 1 (50 pts):

Convolutional Neural Networks (CNNs) are a class of deep neural networks commonly used to analyze visual imagery. In this problem, you are tasked to leverage CNNs to classify traffic signs, which is crucial in self-driving cars.

1. **(5 pts)** Explain the architecture of AlexNet Convolutional Neural Network and indicate the use of each layer, such as convolution/feature extraction, max pooling, and flattening layers.
2. **(5 pts)** Explain the concept of overfitting and underfitting and suggest techniques to overcome them in CNNs.
3. **(10 pts)** Visualize the dataset and count how many samples of images belong to each class.
4. **(10 pts)** Train a CNN to classify traffic signs using only 15 epochs. Plot the network accuracy on the testing dataset.
5. **(10 pts)** Explore 3 different ways to improve the network accuracy, such as increasing the number of epochs, adding additional convolutional/max-pooling layers, and adding dropout layers. Retrain the network and plot the results. Any reasonable answer should be sufficient.
6. **(10 pts)** Build 3 different deep convolutional neural networks with various architectures. Feel free to choose the number of layers, kernels (feature detectors), dense (Fully Connected) layers, neurons in the dense layer, activation functions, regularization such as dropout, etc. Retrain the network and plot the results.

Problem 2 (50 pts):

In this problem, you are tasked with developing an artificial neural network (ANN) to solve the Happy House Problem, where only individuals with smiling faces will be allowed entry. The goal is to create a robust smile detection model using a dataset of images, where each image is a 64x64 colored image with pixel values ranging from 0 (black) to 255 (white). Your ANN will be trained to identify whether a person is smiling or not based on the provided images. The dataset consists of labeled images of faces, and your task is to build, train, and evaluate a model that can accurately classify smiling faces. The solution must aim for a high accuracy of at least 80% on the testing datasets.

1. **(10 pts) Visualize the dataset and perform basic exploratory analysis**:
   - Display some sample images of faces from the dataset (both smiling and not smiling).

- Count the number of smiling and non-smiling images in the dataset to ensure a balanced or imbalanced understanding of the dataset.

2. **(10 pts) Build and train an ANN for smile detection**:
   - Construct a basic ANN model that processes the 64x64 colored images to classify if a person is smiling.
   - Train the model for a set number of epochs (e.g., 20 epochs) and plot the network's accuracy on the testing dataset.
3. **(10 pts) Evaluate the model and explore ways to improve performance**:
   - Evaluate the model's performance on the test dataset by plotting accuracy and loss over epochs.
   - Suggest and explore at least two techniques to improve the model's accuracy, such as increasing the number of epochs, adjusting the number of hidden layers, or adding dropout layers for regularization.
4. **(20 pts) Build alternative ANN architectures and compare results**:
   - Build two additional variations of the ANN architecture by modifying elements like the number of hidden layers, activation functions, or number of neurons per layer.
   - Retrain the networks, plot the results, and compare each model's accuracy and loss performance.